B NAGARAJU REDDY

Platform Engineer | Site Reliability Engineer | Certified Kubernetes Administrator (CKA)

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Summary

Senior Software Developer specializing in Platform Engineering with 3 years of end-to-end infrastructure ownership at ZeOmega, a HIPAA-compliant healthcare SaaS. Independently designed and delivered production-grade HA systems (Redis Sentinel 3+3, Memcached HA), full mTLS security architecture, a 20-component observability stack, and Jenkins CI/CD pipelines that cut deployment cycles from 3 weeks to under 24 hours. Certified Kubernetes Administrator (CKA, 2026) with advanced bare-metal cluster experience covering networking, persistent storage, and production incident resolution. Strong in performance engineering, security hardening, and reliability operations.

Experience

- **ZeOmega** — Healthcare SaaS, HIPAA-compliant platform *July 2023 – Present*
  Senior Software Developer (Platform Engineering) Bengaluru, India
  - **Architected Redis Sentinel HA (3+3 topology) and Memcached HA** from scratch across a microservices platform, eliminating all single points of failure; validated with Memtier Benchmark & Locust across throughput, latency, failover time, and memory limits; tuned `maxmemory` and eviction policies; documented failover behaviour and Memcached internals.
  - **Built one-click Jenkins CI/CD pipelines** for Redis and Memcached deployments (HA + TLS + version management + integrated testing) — cutting each rollout cycle **from 3 weeks to under 24 hours**; automated x.509 certificate generation and rotation across all pipeline stages.
  - **Introduced mTLS security architecture from scratch** across inter-service communication using x.509 certificates; architected a highly available HashiCorp Vault cluster with AppRole authentication and automated certificate rotation; enforced Redis multitenancy via ACL-based data isolation and delivered audit logging for HIPAA compliance.
  - **Independently built end-to-end observability platform** using Prometheus + Grafana covering **20+ infrastructure components** (RabbitMQ, Redis HA, Memcached, MongoDB, RHEL & Windows VMs); configured `mongodb_exporter` telemetry; reported and validated metric gatherer registration conflicts under MongoDB 8.3.x (percona/mongodb_exporter#1285).
  - **Deployed and managed stateful workloads** on a **3-node bare-metal Kubernetes cluster** with Longhorn persistent storage; resolved production DNS caching incidents; stress-tested workloads for fault tolerance and data durability before rollout.
  - **Eliminated production message-drop incidents** by redesigning RabbitMQ batch consumer with optimised processing workflows **— 200% throughput improvement**, zero data loss across critical healthcare data pipelines; directly reduced client-facing downtime.
  - **Benchmarked Milvus vs pgvector** across 1M vectors (1024-dim) for latency, recall & throughput; findings directly influenced team's vector search architecture; contributed bug fix merged into VectorDB-Bench open source codebase.

Projects

- **Smart Queue Processor** **[github]**
  Python · Redis · RabbitMQ · Docker · FastAPI
  - **Fault-tolerant messaging infrastructure:** Backend-agnostic event-driven queue system with pluggable adapter architecture (Redis/RabbitMQ); implemented batching, configurable retry logic, and Dead Letter Queue (DLQ) for guaranteed message delivery and zero data loss under failure conditions; Dockerised for consistent multi-environment deployment; FastAPI load producer enables throughput benchmarking and stress testing.

Technical Skills

- **Orchestration & Containers:** Kubernetes (bare-metal, CKA 2026), Docker, Helm, Longhorn persistent storage, Samba
- **Cloud & Infrastructure as Code:** AWS (EC2, VPC, IAM, S3, Route53), Terraform
- **CI/CD & Automation:** Jenkins, SCM-based pipelines, x.509 cert automation, one-click rollout pipelines
- **Observability & SRE:** Prometheus, Grafana, mongodb_exporter, structured logging, automated alerting
- **Messaging & Caching:** RabbitMQ (HA clustering, batch processing), Redis (Sentinel HA, 3+3 topology), Memcached HA
- **Security & Compliance:** HashiCorp Vault (HA, PKI, AppRole), mTLS, x.509 Certificates, ACL multitenancy, HIPAA compliance
- **Performance Testing:** Memtier Benchmark, Locust, k6, failover validation, capacity planning
- **Languages & Databases:** Python, Bash · MongoDB, PostgreSQL, MySQL, Vector DBs (Milvus, pgvector)
- **Operating Systems & Networking:** Linux Administration (RHEL, Ubuntu), DNS, TLS/mTLS, TCP/IP fundamentals

Education

- **IIIT RGUKT — B.Tech, Electronics and Communication** *Aug 2019 – May 2023*
  CGPA: 8.93 / 10.00 RK Valley, India

Certifications

- **Certified Kubernetes Administrator (CKA)** — The Linux Foundation, Feb 2026 [verify]
- The Complete Data Structures and Algorithms in Python — Udemy, Dec 2022 [certificate]